Enzora
User Guide

Mutation-library construction, evidence-informed screening and candidate delivery



A practical guide to the current web workspace

Start with your enzyme and substrate. Build a candidate library, compare predicted properties, then review a shortlist for the next experiment.

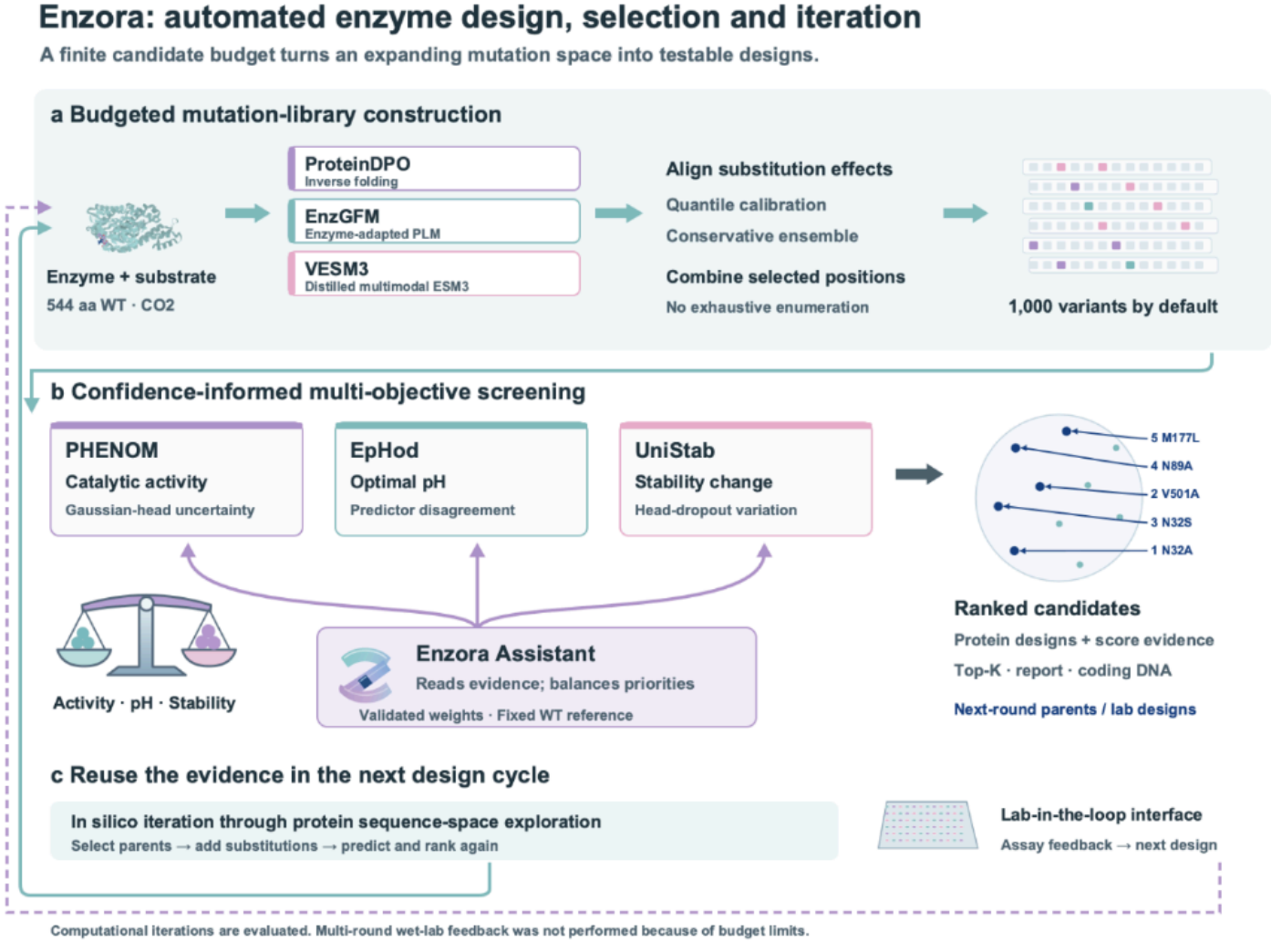
Website

https://enzora-agent.com

For researchers using the browser, follow Sections 1–13.
For local installation and model resources, see Section 14.
For recovery and interpretation checks, see Sections 15–16.

1. The three-stage workflow

Each stage has a clear input, action and saved result



Library: Build library

Supply the WT protein and substrate, then choose the mutation budget.

Screening: Start Screening

Run the three predictors. Review priorities and uncertainty before selection.

Output: Select Top-K & generate report, then inspect and export.

2. Access and Demo mode

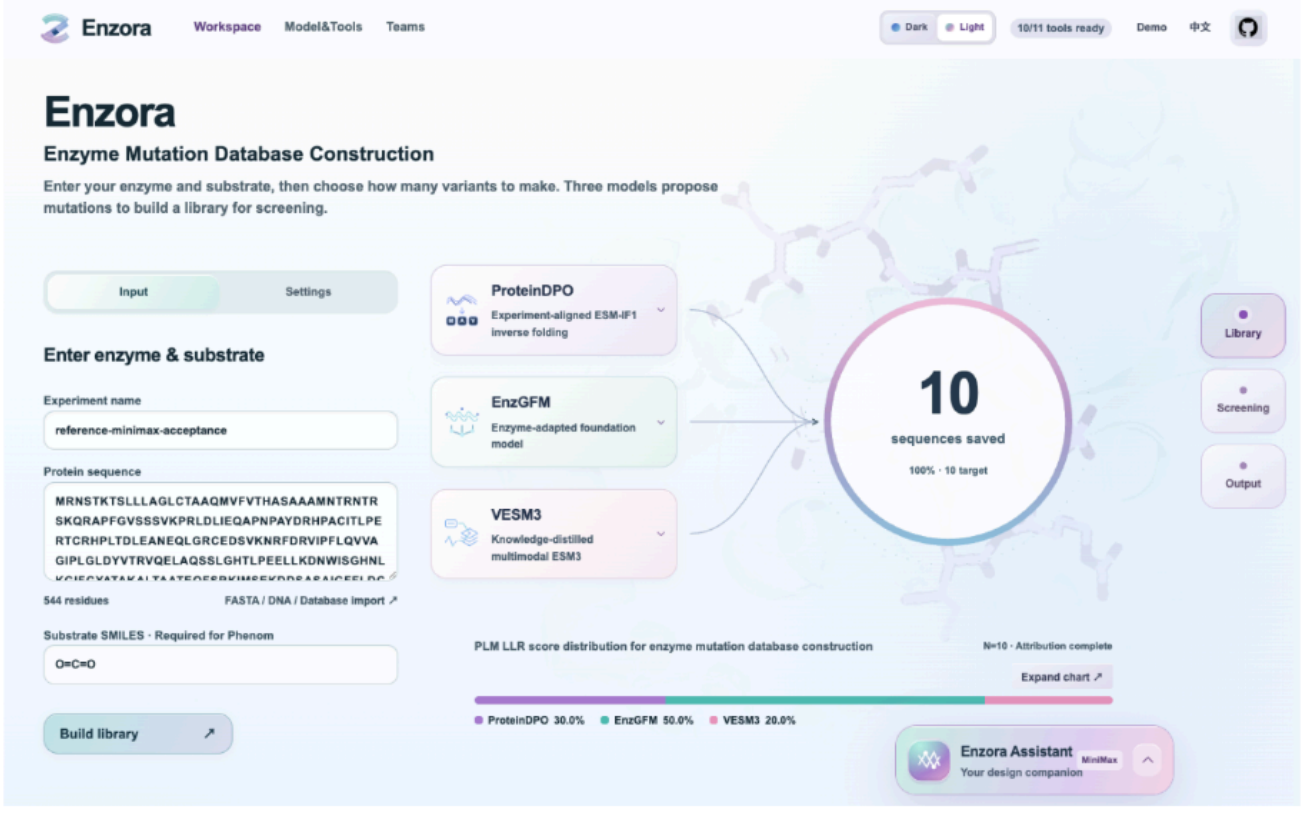
The access code opens the workspace. Provider credentials stay on the server

Open the production platform

Visit enzora-agent.com. Enter the workspace access code when requested. The access code is separate from the MiniMax API credential.

Try the interface before submitting a real job

The Demo toggle runs local simulated data. Its fixed example is the project's 544 aa carbonic anhydrase with CO2, entered as O=C=O. Demo predictions and its projection are simulated.



Scrolling changes the view. It does not authorize a new computation.

3. Enzyme and substrate inputs

Check the protein identity before starting a run

Protein sequence

Paste a canonical amino-acid sequence or use FASTA / DNA / Database import. The joint workflow accepts 1–1,000 residues. Translate DNA/RNA and confirm the protein before submission.

Substrate SMILES

PHENOM needs an enzyme–substrate pair. For the carbonic anhydrase example, carbon dioxide is O=C=O. A pH target alone does not replace the substrate.

Files and the unified object

The import route accepts FASTA, CSV, BioArtifact JSON, GenBank and sequence-bearing SBOL. Review the parsed records and choose the intended protein. The file-entry limit is 100 records.

Structure correspondence

The inverse-folding path needs the configured structure and chain to match the WT. Use a matching structure or ask the administrator to verify the mapping before a new enzyme is run.

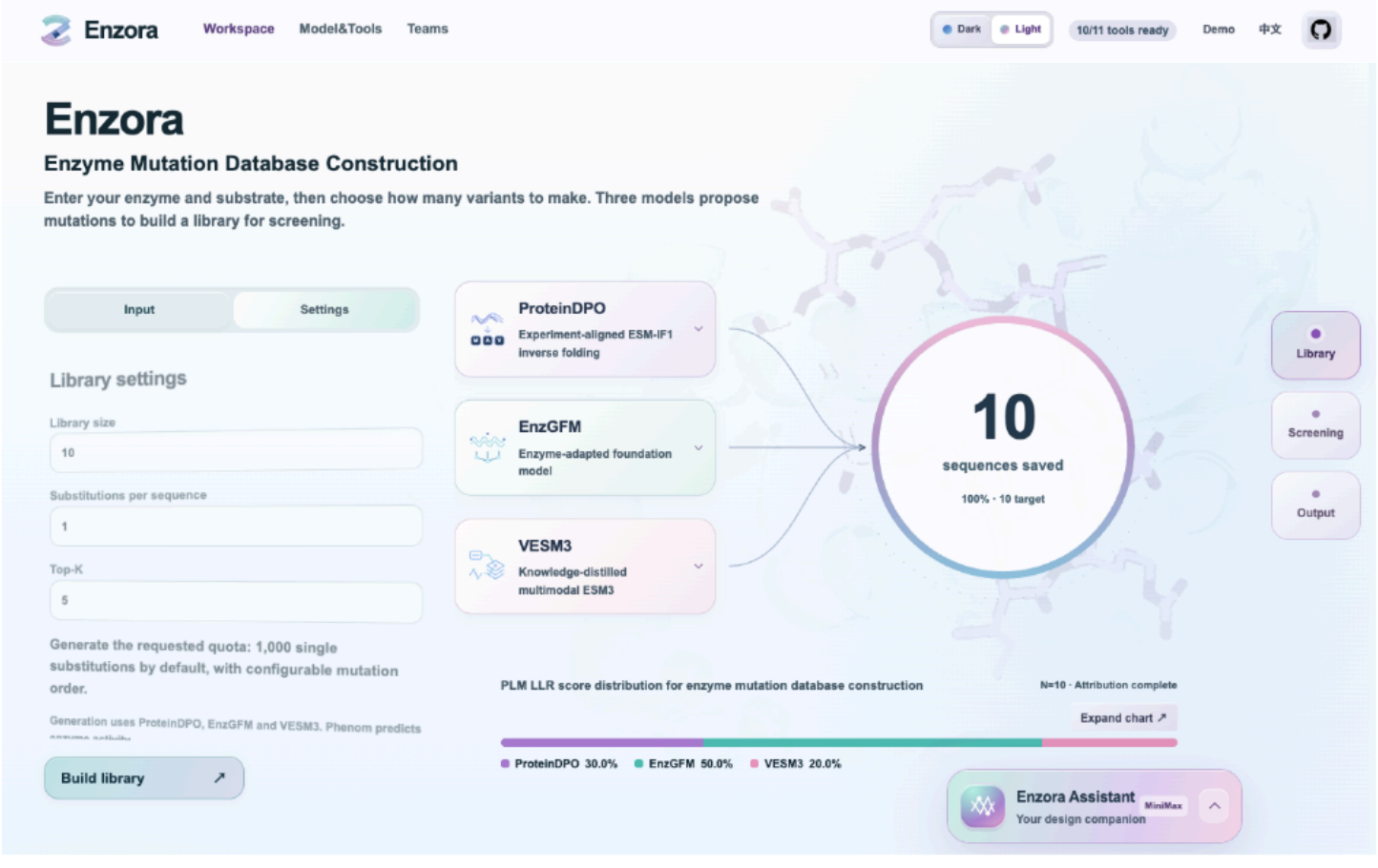
Keep the original protein identifier. It links mutations, model predictions and exported files.

4. Library settings

The budget controls how many unique variants enter screening

Budget, mutation order and Top-K

The default library contains 1,000 single mutants, with five final candidates. The Settings tab changes library size, mutation order and Top-K. A double mutant changes two distinct residues.



Advanced restrictions

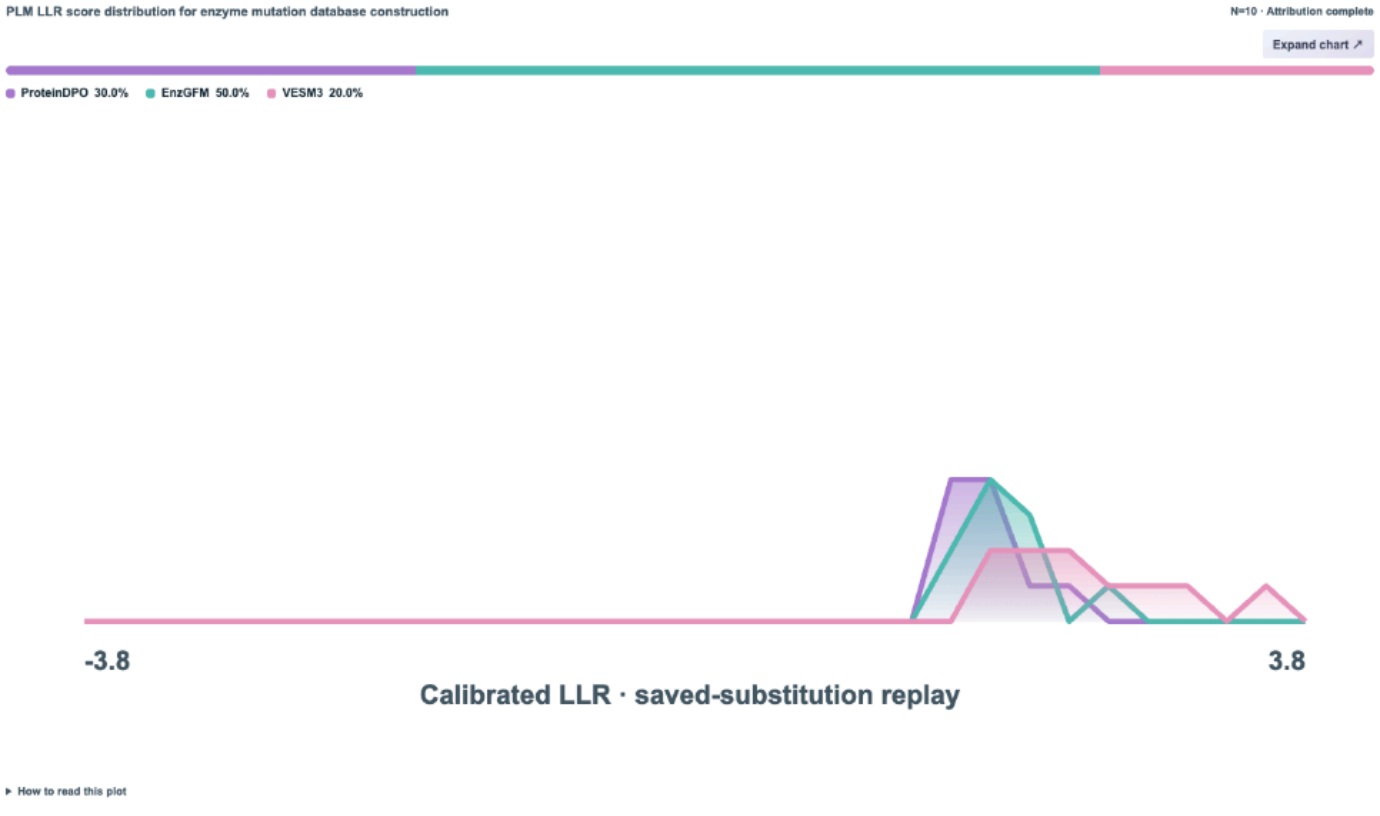
Allowed/excluded positions and mutation-order quotas are request fields. Review them in the preparation/parameter preview when used. Quotas must sum to the library size.

5. Construction progress and LLR

Saved records and model scores answer different questions

The circular counter

The ring tracks saved candidate sequences, not a benchmark of inference speed. Library completion enables Start Screening. Wait for completion before requesting the next computation.



The LLR distribution and attribution bar

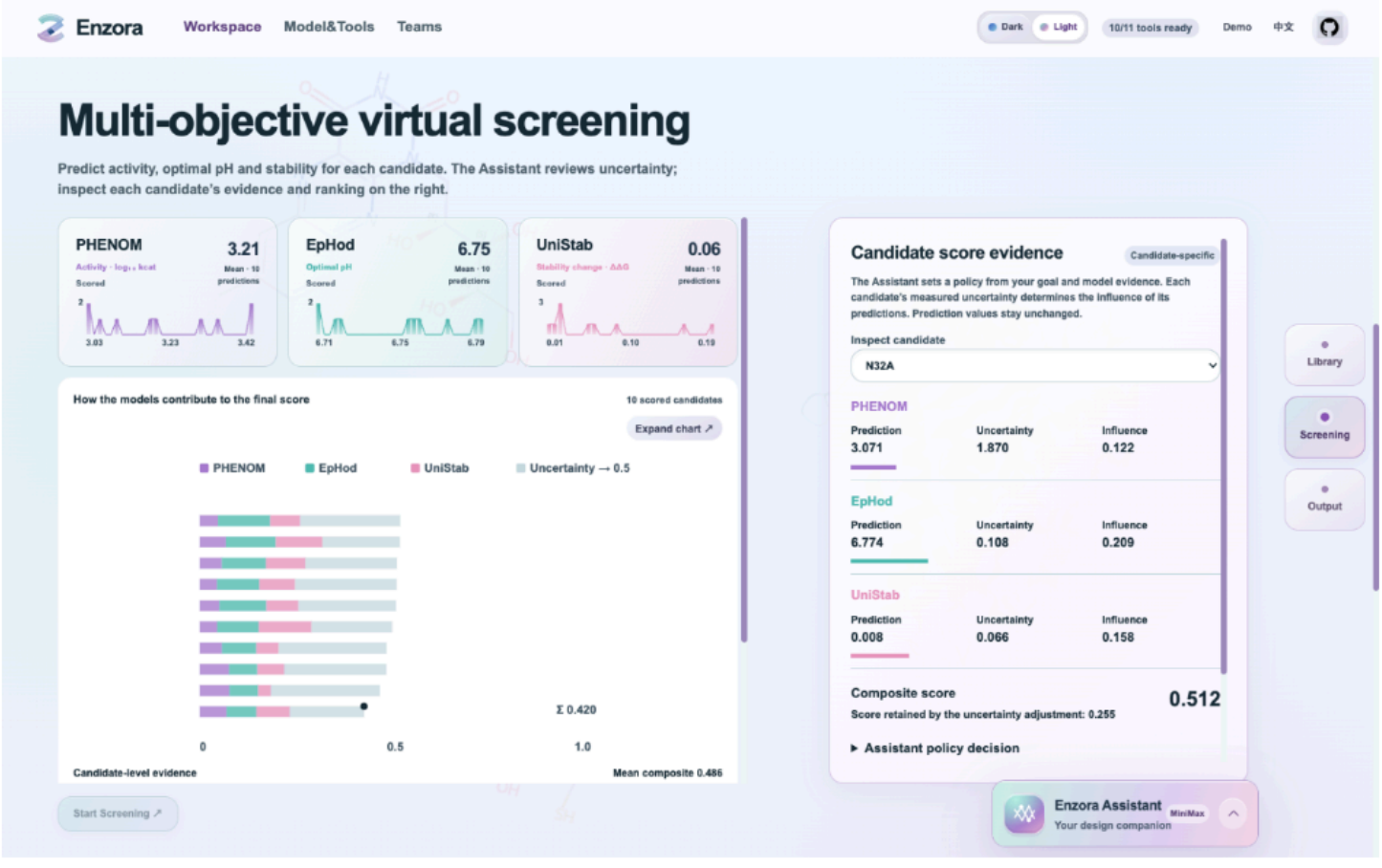
The curves show calibrated model scores for saved substitutions. The bar assigns each substitution to its lowest-scoring model; ties share credit. The five-second animation replays saved data.

6. Virtual screening

Three predictors provide distinct biological quantities

PHENOM, EpHod and UniStab

PHENOM predicts log10(kcat) and a Gaussian standard deviation. EpHod predicts optimal pH and reports predictor disagreement. UniStab predicts WT-relative stability change and head-dropout variation.



Start Screening is an execution boundary

Scroll to Screening and click Start Screening after the library is ready. Queue time and model execution are separate states. Missing outputs are not zero scores.

7. Candidate score evidence

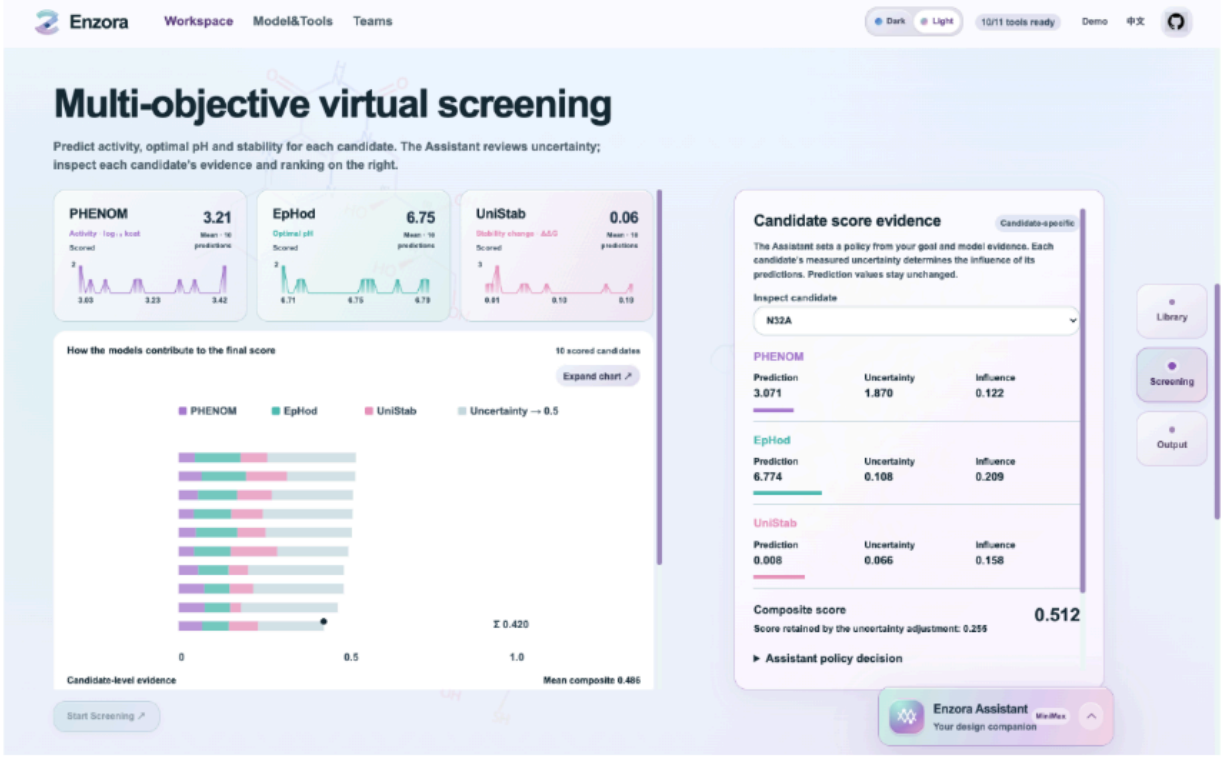
Inspect how uncertainty affects each candidate, without fixed percentage controls

The Assistant sets an evidence-informed policy

Choose a candidate in the right panel. Compare its model predictions, measured uncertainty and actual influence coefficients. The Agent sets a reliability policy; code applies uncertainty using each model's own scale. Predictions remain unchanged.

Review the applied balance

Open Assistant policy decision to inspect the recorded tool explanation. The chart separates measured prediction evidence from the neutral baseline retained under uncertainty. Continue to Output when scores are ready.



Confidence is a control parameter, not an experimental success probability.

8. Reading the score

A fixed WT reference makes gains interpretable

How model evidence becomes a candidate score

Global priorities and candidate-specific uncertainty play different roles.

a Global priorities

$$W_i = \frac{P_i C_i}{\sum_{j=1}^{3} P_j C_j}$$

Preference sets the importance of each property.
Confidence changes how much a predictor influences ranking.
The three effective weights sum to one.

b Candidate-specific uncertainty

$$r_{k,i} = \frac{1}{1 + u_{k,i}/\tau_i}$$

Larger uncertainty reduces the size of a predicted gain or loss.

$$A_{k,i} = \frac{1}{2} + r_{k,i}\left(\frac{1}{1 + e^{-g_{k,i}/s_i}} - \frac{1}{2}\right)$$

Gain compares the candidate with WT. Scales stay frozen.

c Final candidate score

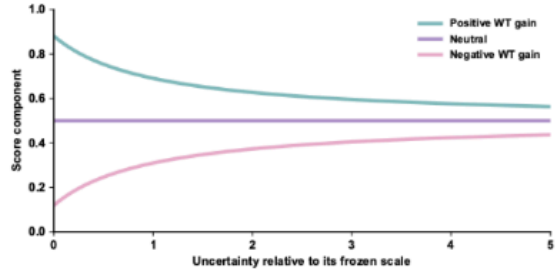
$$S_k = \sum_{i=1}^{3} W_i A_{k,i}$$

Combine the three normalized components.

Threshold filters remain separate from ranking.

Analytical explanation of the implemented policy, not a calibration result.

d Increasing uncertainty softens the evidence



Gain, uncertainty and the neutral point

Activity gain is candidate minus WT. pH gain is the reduction in distance to the target. Stability gain follows the configured lower-is-better ΔΔG convention. Uncertainty shrinks gains and losses towards 0.5.

Ranking and acceptance are separate

A high rank does not mean that a candidate meets the target. Review pH tolerance, minimum activity and stability thresholds alongside the composite score.

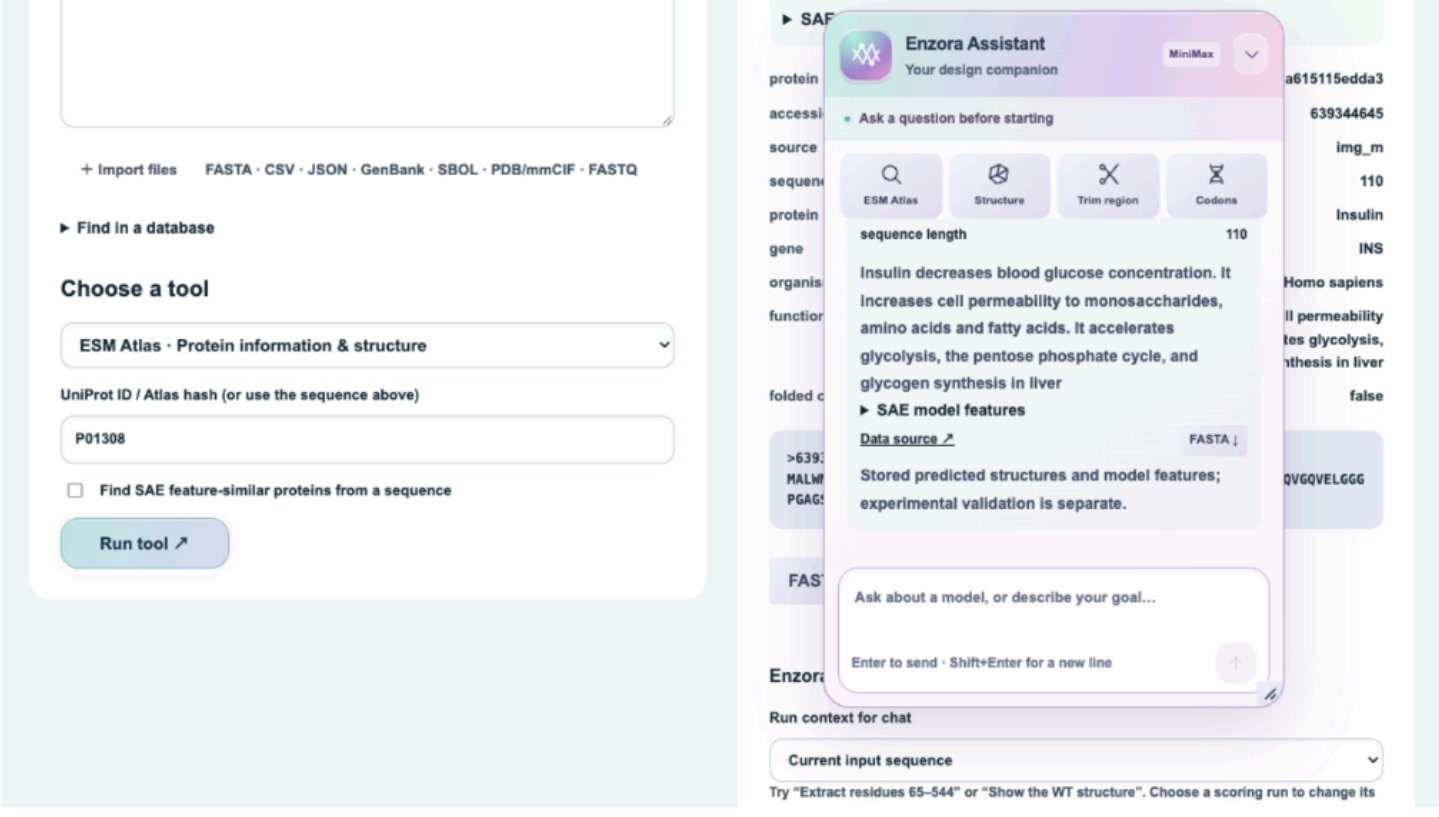
9. Enzora Assistant

Natural language requests call bounded, validated tools

Context and progress

Open the floating Assistant, drag its header or resize its edges. Follow the named tool steps and elapsed time during a request. Recent conversation turns help preserve context.

“Explain why this candidate ranks first.”
“Look up UniProt P01308 in ESM Atlas.”
“Extract residues 25–300 from the supplied protein.”



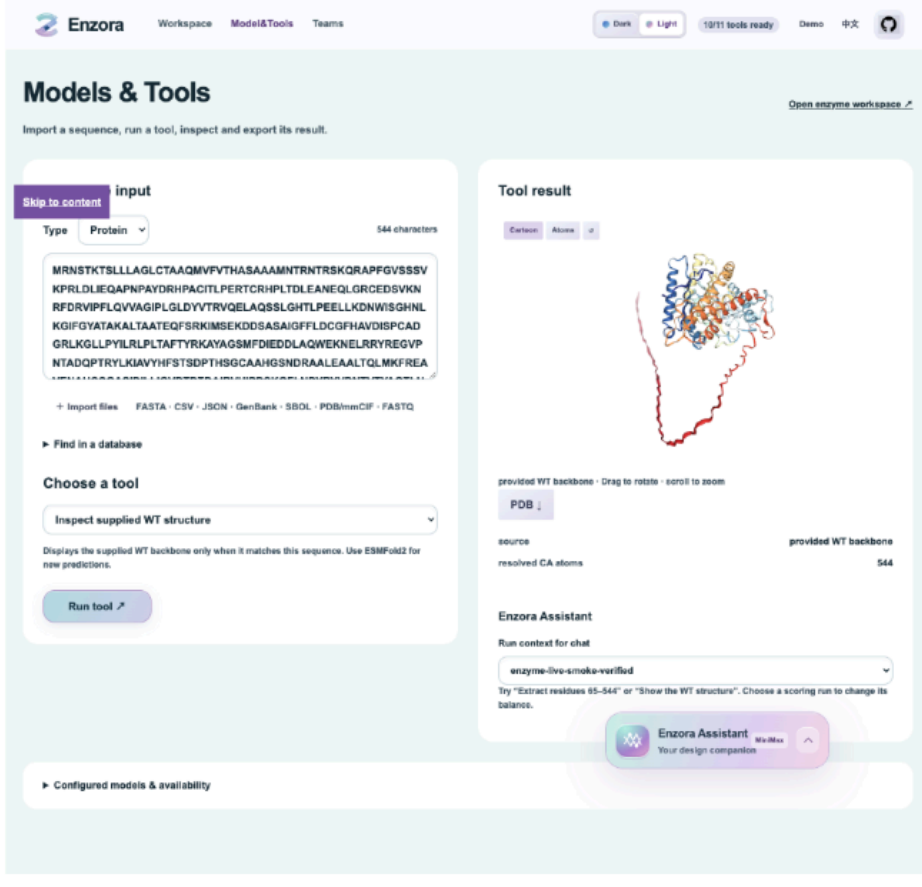
A residue interval must come from your instruction. The tool does not infer a cleavage site.

10. Structures and Model&Tools

Inspect the source and identity of every returned structure

ESM Atlas and the structure viewer

Atlas lookup retrieves metadata, sequence, SAE features and any stored prediction. The viewer supports rotation, zoom and representation changes. SAE feature similarity is not sequence identity.

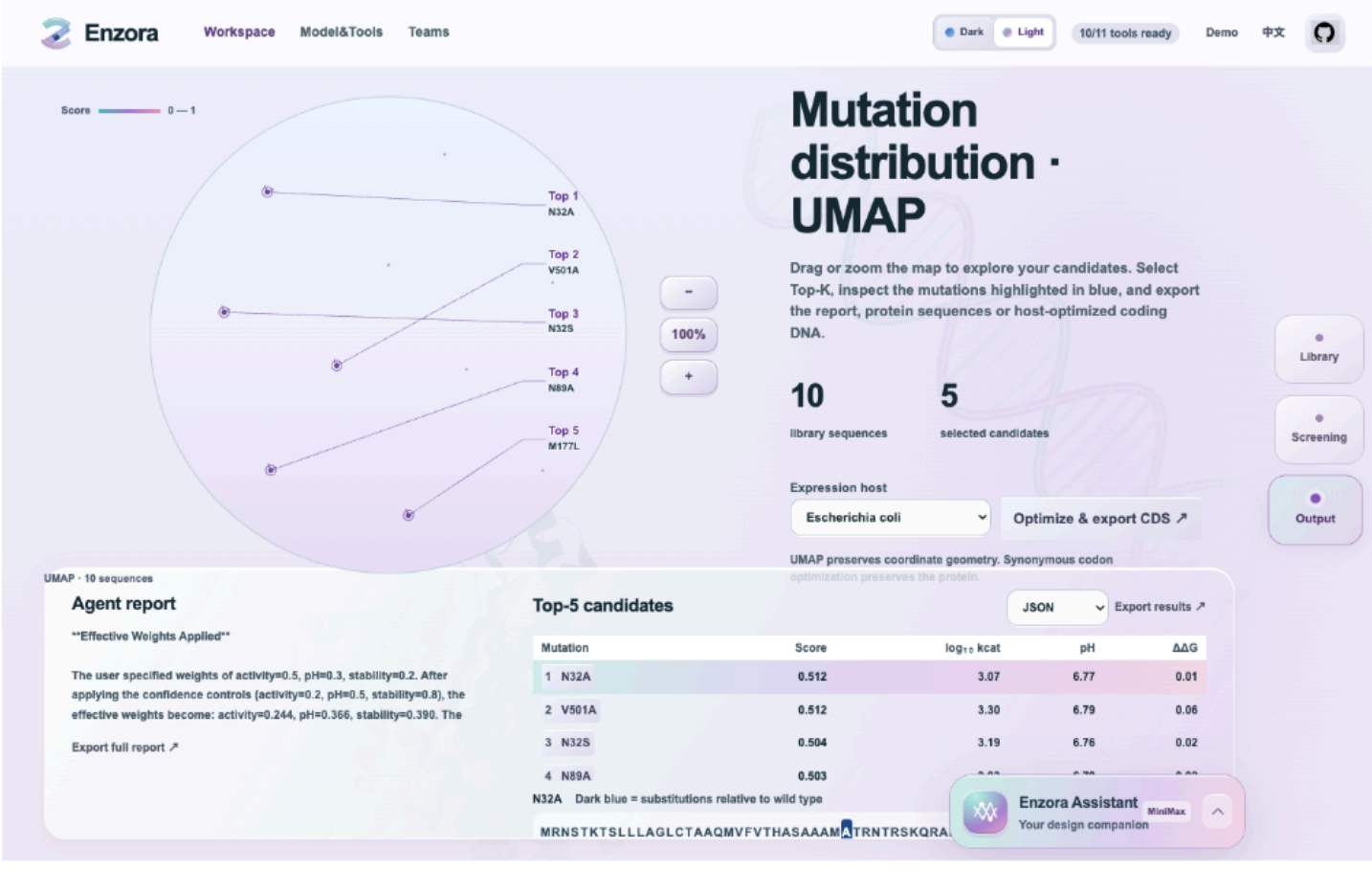


Prediction is a separate request

Inspect structure displays the supplied WT structure. ESMFold2 prediction requires a separately configured Biohub token and an explicit request. The Tools page disables unavailable operations.

11. UMAP and the Top-K shortlist

The map helps you inspect sequence space, not measure experimental activity



Explore and select

Drag the map to pan and use the wheel or zoom buttons. Select a highlighted Top-K node to inspect its sequence. The dark-blue residues identify substitutions relative to the initial WT.

Generate the final result

After scoring is ready, go to Output and click Select Top-K & generate report. This starts final selection, UMAP and analysis. The report export becomes available when those steps finish.

12. Result formats and coding DNA

Choose a format that preserves the information you need

Proteins and prediction records

The result selector exports JSON, CSV, FASTA, GenBank or SBOL. JSON preserves the BioArtifact schema, predictions and provenance. FASTA carries sequence identity but does not preserve arbitrary prediction annotations.

Host-specific synonymous DNA

Choose the expression host, then use Optimize & export CDS. The export includes WT and selected candidates. Translate each DNA sequence back to the source protein before gene synthesis.

What codon optimization does

Eleven host usage tables are built in, including E. coli and Bacillus subtilis. The current method chooses preferred synonymous codons. It does not optimize mRNA folding, restriction sites, repeats or synthesis constraints.

Keep a traceable package

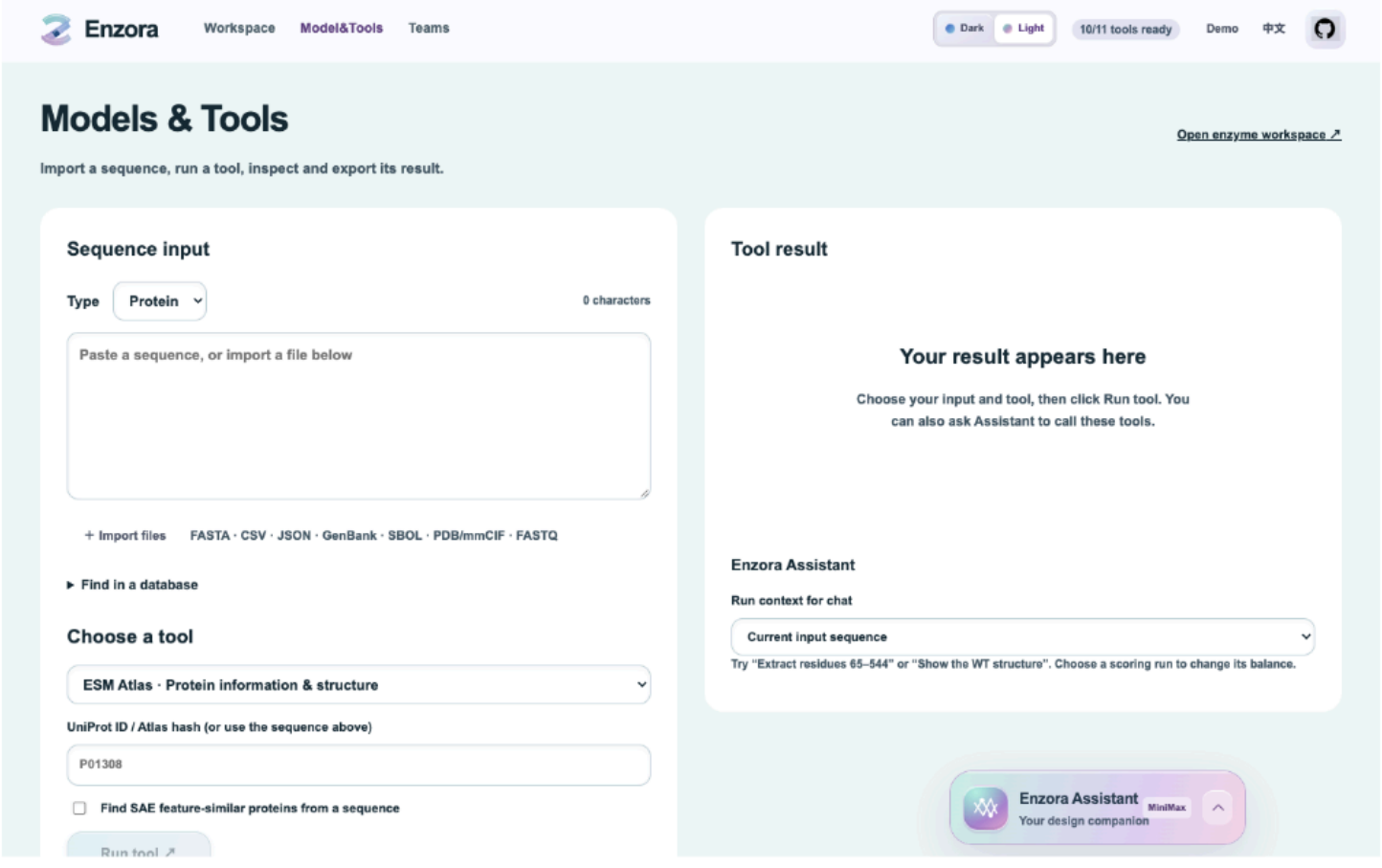
Save the protein FASTA together with JSON/CSV and the analysis report. Check candidate identifiers, sequence lengths, substitutions, substrate and host across all exported files.

13. Model&Tools operations

Use a specific tool without starting the full mutation workflow

Choose the input and operation

Open Model&Tools. Enter a protein sequence or accession, choose the operation and inspect its returned card. Atlas queries include source links and sequence downloads. An active scientific run is not silently replaced.



Understand availability

Structure inspection, explicit sequence trimming and host codon conversion use the configured tools. External prediction needs its own resource setup. A disabled tool is not evidence of a completed calculation.

14. Local setup commands

Run commands from the repository root

```
git clone https://github.com/YuzhangLi2281/Enzora.git
cd Enzora
python3 -m venv .venv
source .venv/bin/activate
pip install -r src/Enzora_Software_Webs/utils/server/requirements.txt
npm --prefix src/Enzora_Software_Webs/web ci
```

Configure resources before launching

Edit config/server.yaml and config/mutation.yaml. Set model interpreters, repositories and weight paths. Configure MiniMax privately and run the path checks before submitting a real task.

API terminal

```
python src/Enzora_Software_Webs/utils/server/run.py \
  --host 127.0.0.1 --port 8000
```

Frontend terminal

```
VITE_PROXY_TARGET=http://127.0.0.1:8000 \
  npm --prefix src/Enzora_Software_Webs/web run dev
```

14. Local deployment and resources

For administrators and developers

Code and model environments

Clone the Enzora repository and install its API/web dependencies. Follow each model's environment requirements. PHENOM inference also requires its checkpoint, ESMC and MolT5 resources. Enzora does not distribute all model weights.

Configuration entry points
config/server.yaml
config/mutation.yaml
config/request.example.yaml

Provider: MiniMax-M2.5
Base URL: https://api.minimax.cn/v1

Private credentials

Keep MINIMAX_API_KEY and the workspace access token in the server's private environment/configuration. Never embed them in config.js, frontend code, screenshots or public repository history.

Execution and cache boundaries

The API submits model jobs to the configured scheduler. Shared weights remain separate from per-run data. Success, failure and cancellation clear disposable graphs/logits while retaining predictions, reports and audit records.

15. Recovery and interpretation checks

A completed interface action does not guarantee a completed model job

Login fails

Use the workspace access code, not the MiniMax provider key.

A job remains queued

Check the job status. Repeated submissions do not accelerate the current allocation.

A model fails

Read the stage error and retained logs. Verify its environment, weights and input contract.

The Assistant takes time

Watch actual tool progress. Network/provider delays can outlast a scientific tool call.

No candidate meets the target

Review acceptance filters separately from rank. Revise the design or use a new round.

The report is unavailable

Wait for selection/analysis to finish. A failed report is not an empty successful result.

A run is cancelled

Confirm the scheduler has stopped the job before cleaning active files.

16. Evidence and further reading

The guide follows the current browser and production code

Verified integration example

The screenshots use a real 544 aa WT / CO2 task with ten single-mutant candidates and Top-5 output. All three generators, all three scorers and MiniMax native balancing completed. This is an integration check, not 1,000-candidate performance or wet-lab validation.

Scientific interpretation

UMAP uses positional amino-acid identity with Hamming distance. It is a visualization. Model uncertainty proxies have different meanings and are not jointly calibrated success probabilities. Predicted candidates require experimental testing.

Further reading
https://enzora-agent.com
https://github.com/YuzhangLi2281/Enzora
https://github.com/YuzhangLi2281/PHENOM
https://github.com/ntranoslab/vesm
https://github.com/ai4protein/VenusFactory2

Project diagrams retain editable labels and vector components. Native formula sources remain available for editing.